Synopsis

Air Quality Level Classification System

Problem Id: 19

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Section : L

- **Problem Description:**

Air pollution is increasing rapidly due to industrialization, vehicles, and urban development. Continuous monitoring of air quality using sensors generates large amounts of environmental data. However, analyzing this data manually is difficult.

The problem is to develop a machine learning model that can classify air quality levels based on sensor readings such as CO, NO_2, temperature, and humidity. The system should preprocess the data, select relevant features, and apply classification algorithms to accurately determine air quality.

Air pollution is one of the major environmental issues affecting human health and ecosystems. Monitoring and predicting air quality helps authorities take preventive actions. This project proposes a machine learning based Air Quality Level Classification System that predicts air quality levels using sensor data such as CO, NO_2, temperature, and humidity.

The dataset is preprocessed to remove noise, handle missing values, and normalize sensor readings. Important features are selected to improve model efficiency. Two machine learning algorithms, Logistic Regression and K-Nearest Neighbors (KNN), are used to classify air quality levels.

The models are evaluated using accuracy and precision metrics to determine their performance.

- **Steps of Implementation:**

1. Data Collection:

First, we will gather an air quality dataset that includes parameters like CO, NO_2, temperature, and humidity. This data can be taken from platforms like Kaggle or other open data sources.

Feature Name	Description	Data Type	Possible Values
CO (Carbon Monoxide)	Concentration of CO in air	Float	0.1 – 10.0 ppm
NO_2 (Nitrogen Dioxide)	Concentration of NO_2	Float	5 – 200 µg/m³
SO_2 (Sulfur Dioxide)	Air pollutant level	Float	2 – 150 µg/m³
O_3 (Ozone)	Ozone concentration	Float	10 – 180 µg/m³
PM2.5	Fine particulate matter	Float	0 – 500 µg/m³

Feature Name	Description	Data Type	Possible Values
PM10	Coarse particulate matter	Float	0 – 600 µg/m³
Temperature	Ambient temperature	Float	0 – 50 °C
Humidity	Relative humidity	Float	10 – 100 %
Wind Speed	Speed of wind	Float	0 – 20 m/s
Wind Direction	Direction of wind	Categorical	N, S, E, W, NE, NW, SE, SW
Timestamp	Date & time of observation	Datetime	2026-03-28 10:30:00
AQI (Target Variable)	Air Quality category	Categorical	Good, Moderate, Poor, Very Poor, Hazardous

2. Data Preprocessing:

Next, we will clean the data by handling missing values and removing unwanted noise. We will also scale the data so that all values are in a similar range, which helps the model perform better.

To improve model performance, the following preprocessing steps will be applied:

- **Handling Missing Values**
  - Use mean/median imputation for numerical data
  - Use mode for categorical data
- **Noise Removal**
  - Remove outliers using IQR (Interquartile Range) or Z-score method
- **Feature Scaling**
  - Apply **Min-Max Scaling** or **Standardization (Z-score normalization)**
  - Important for distance-based models like KNN
- **Encoding Categorical Data**
  - Convert wind direction or AQI categories using **Label Encoding / One-Hot Encoding**
- **Feature Selection**
  - Use correlation matrix or feature importance to remove less useful features

3. Model Selection:

We will use the following algorithms:

K-Nearest Neighbors (KNN)

- Works well for classification problems
- Uses distance between data points
- Suitable because air quality data has continuous numerical features

Logistic Regression

- Simple and efficient classification algorithm
- Provides probabilistic outputs
- Works well when relationship is linear

4. Training and Validation:

- The dataset will be split into:
    - **Training Set (70–80%)**
    - **Testing Set (20–30%)**
- Training Process:
    - Models (KNN & Logistic Regression) will be trained using training data
- Validation Techniques:
    - **Train-Test Split Method**
    - **Cross-Validation** for better reliability
- Hyperparameter tuning:
    - K value in KNN
    - Regularization in Logistic Regression

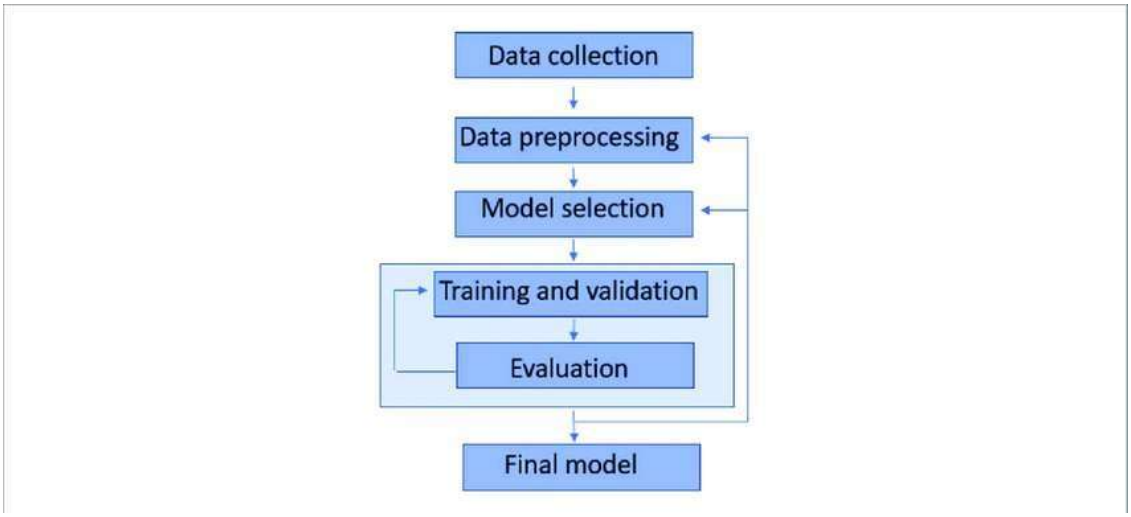
5. Evaluation:

The performance of models will be evaluated using:

- **Accuracy** → Overall correctness
- **Precision** → Correct positive predictions
- **Confusion Matrix** → Detailed classification results
- *(Optional)*: Recall & F1-score for deeper analysis

6. Final Model:

Finally, we will select the model that gives the best results and use it for predicting air quality levels.



- **Expected Modules/ Libraries to be used :**

1. NumPy:

It is used to perform numerical calculations and work with arrays efficiently, which makes handling large amounts of data easier.

2. Pandas:

It helps in organizing and managing the dataset, making it simple to read, clean, and analyze data.

3. Matplotlib:

It is used to create basic graphs and charts, helping us visualize the data in a clear way.

4.Seaborn:

It provides more advanced and attractive visualizations, such as heatmaps, to better understand relationships in the data.

5. Scikit-learn (sklearn):

It is the main library used to build machine learning models like KNN and Logistic Regression, and also to evaluate their performance using different metrics.

- **Expected Evaluation metrics and Plots to be used:**

1. Accuracy: Measures overall correctness of the model.

2.Precision: Measures correctness of positive predictions.

3.Confusion Matrix: Shows correct and incorrect classifications.

4. Heatmap: Visual representation of correlation between features.

5. Bar Graphs: Used to compare model performance.

These metrics and plots help in understanding model performance and selecting the best model.

- **Required topics from the subject:**

1. Data Structures (Arrays):

Arrays are used to store and manage the dataset values in an organized way, making it easier to process and perform operations on the data.

2. File Handling:

File handling is used to read the dataset (usually in CSV format) so that it can be used in the program for further processing.

3. Functions:

Functions help in breaking the code into smaller, manageable parts, which improves readability and allows code reuse.

4. Basic Python Programming:

Basic Python concepts are used to implement the overall logic of the project and connect all the steps together smoothly.

- **Platform Required:**

1. Jupyter Notebook
2. Python IDLE
3. Visual Studio Code

- **Books and Link Sources:**

1. "Hands-On Machine Learning with Scikit-Learn" by Aurélien Géron
2. Python for Data Analysis" by Wes McKinney
3. Kaggle (https://www.kaggle.com/□)
4. Scikit-learn Documentation (https://scikit-learn.org/□)